EXAM QUESTION

0 → not a zero
• → just a component

h_t = $\begin{bmatrix} \bullet \\ \bullet \\ \bullet \\ \bullet \end{bmatrix}_{4\times 1}$ => This is answer (left side)

W_{xh} = $\begin{bmatrix} \bullet & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{4\times 3}$ Another dimension to multiply

$[W_{xh} x_t]$

x_t = $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}_{3\times 1}$

h_{t-1} = $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}_{4\times 1}$

$[W_{hh} h_{t-1}]$

W_{hh} = $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{4\times 4}$

Input will be size of the length of the dictionary. which is $n \times 1$

$$h_t = \tanh\left(\underset{3\times 3}{W_{hh}}\ \underset{3\times 1}{h_{t-1}} + \underset{3\times 2}{W_{xh}}\ x_t\right)$$

$x^{(1)}$: $x_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ Time step 1 $\quad x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ Time step 2

Input dimension = 2 $\quad x_t \in \mathbb{R}^2$

hidden state dim = 3 $\quad h_t \in \mathbb{R}^3$

output dimension = 2 $\quad y_t \in \mathbb{R}^2$

Initial hidden state = h_0 = $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}_{3\times 1}$

W_{xh} = 3x2

W_{hh} = 3x3

W_{xy} = 2x3

W_{hy} = 2x3

$$W_{xh} = \begin{bmatrix} 0.5 & -0.3 \\ 0.8 & 0.2 \\ 0.1 & 0.4 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$W_{hh} = \begin{bmatrix} 0.1 & -0.4 & 0.0 \\ -0.2 & 0.3 & 0.2 \\ 0.05 & -0.1 & 0.2 \end{bmatrix}$$

$$W_{hy} = \begin{bmatrix} 1 & -1 & 0.5 \\ 0.5 & 0.5 & -0.5 \end{bmatrix}$$

$$h_t = \tanh\left(W_{hh}\, h_{t-1} + W_{xh}\, x_t\right)$$

$h_1 = \tanh$

While $t = 1$

$$W_{hh}\, h_{t-1} = \begin{bmatrix} 0.1 & -0.4 & 0.0 \\ -0.2 & 0.3 & 0.2 \\ -0.05 & -0.1 & 0.2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$3 \times 3 \qquad 3 \times 1$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}_{3 \times 1}$$

$$W_{xh}\, x_t = \begin{bmatrix} 0.5 & -0.3 \\ 0.8 & 0.2 \\ 0.1 & 0.4 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 1 \\ 2 \end{bmatrix}_{2 \times 1} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 - 0.6 \\ 0.8 + 0.4 \\ 0.1 + 0.8 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} -0.1 \\ 1.2 \\ 0.9 \end{bmatrix}$$

$$\begin{bmatrix} -0.1 \\ 1.2 \\ 0.9 \end{bmatrix} \quad \begin{bmatrix} -0.1 \\ 1.2 \\ 0.9 \end{bmatrix}$$

while $t = 2$.

$h_2 = \tanh$

$W_{hh} h_{t-1} = \begin{bmatrix} 0.1 & 0.4 & 0.0 \\ -0.2 & 0.3 & 0.2 \\ -0.05 & -0.1 & 0.2 \end{bmatrix}_{3\times3} \begin{bmatrix} -0.1 \\ 1.2 \\ 0.9 \end{bmatrix}_{3\times1}$

$= \begin{bmatrix} -0.01 + 0.48 + 0 \\ 0.02 + 0.36 + 0.18 \\ 0.005 - 0.12 + 0.18 \end{bmatrix}$

$= \begin{bmatrix} 0.47 \\ 0.56 \\ 0.065 \end{bmatrix}$

$W_{xh} x_t = \begin{bmatrix} 0.5 & -0.3 \\ 0.8 & 0.2 \\ 0.1 & 0.4 \end{bmatrix}_{3\times2} \begin{bmatrix} -1 \\ 1 \end{bmatrix}_{2\times1}$

$= \begin{bmatrix} -0.5 - 0.3 \\ 0.8 + 0.2 \\ -0.1 + 0.4 \end{bmatrix} = \begin{bmatrix} -0.8 \\ 1.0 \\ 0.3 \end{bmatrix}$

$= \begin{bmatrix} 0.47 \\ 0.56 \\ 0.065 \end{bmatrix} + \begin{bmatrix} -0.8 \\ 1.0 \\ 0.3 \end{bmatrix} = \begin{bmatrix} -0.33 \\ 1.56 \\ 0.365 \end{bmatrix}$

$= \tanh \begin{bmatrix} -0.1 \\ 1.2 \\ 0.9 \end{bmatrix} = \begin{bmatrix} -0.09966 \\ 0.833 \\ 0.71 \end{bmatrix}$

$y_t = W_{hy} h_t = \begin{bmatrix} 1 & -1 & 0.5 \\ 0.5 & 0.5 & -0.5 \end{bmatrix}_{2\times3} \begin{bmatrix} -0.09966 \\ 0.833 \\ 0.71 \end{bmatrix}_{3\times1}$

Step 2

$$\begin{bmatrix} -0.09966 - 0.833 + 0.355 \\ -0.04983 + 0.4165 - 0.355 \end{bmatrix} = \begin{bmatrix} -0.57761 \\ 0.01167 \end{bmatrix}$$

while $t = 2$.

$$W_{hh} h_{t-1} = \begin{bmatrix} 0.1 & 0.4 & 0.0 \\ -0.2 & 0.3 & 0.2 \\ +0.05 & -0.1 & 0.2 \end{bmatrix} \begin{bmatrix} -0.09966 \\ 0.833 \\ 0.71 \end{bmatrix}$$

3×3 $\quad$ 3×1

$$= \begin{bmatrix} -0.09966 + 0.3332 + 0 \\ 0.019932 + 0.2499 + 0.142 \\ -0.004983 + 0.0833 + 0.142 \end{bmatrix}$$

$$= \begin{bmatrix} 0.23354 \\ 0.411832 \\ 0.220317 \end{bmatrix} \quad \begin{bmatrix} 0.3833 \\ 0.4117 \\ 0.05375 \end{bmatrix}$$

$$W_{xh} x_t = \begin{bmatrix} 0.5 & -0.3 \\ 0.8 & 0.2 \\ -0.1 & 0.4 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

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$$= \begin{bmatrix} -0.8 \\ 1.0 \\ -0.3 \end{bmatrix}$$

$$\left[W_{hh} h_{t-1} + W_{xh} x_t \right] = \begin{bmatrix} -0.56646 \\ 1.411832 \\ 0.520317 \end{bmatrix} \quad \begin{bmatrix} -0.4767 \\ -0.1883 \\ 0.35375 \end{bmatrix}$$

$$\tanh = \begin{bmatrix} -0.44 \\ -0.18 \\ 0.34 \end{bmatrix}$$

$\rightarrow W_{hy} \times h_t$

$$y_t \Rightarrow \begin{bmatrix} 1 & -1 & 0.5 \\ 0.5 & 0.5 & -0.5 \end{bmatrix} \begin{bmatrix} -0.44 \\ -0.18 \\ 0.34 \end{bmatrix}$$

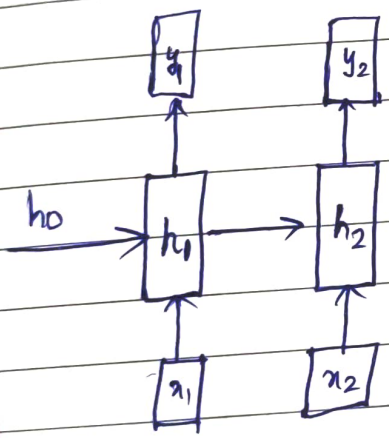
$$y_{t(2)} = \begin{bmatrix} -0.08 \\ -0.48 \end{bmatrix}_{2\times 1} \Rightarrow \text{Step 2.}$$

01/09/2026 Lecture - 29

02/09/2026 Lecture 30, 31

- character level Language Model.
- Backpropagation.

State - H_t (Above problem).



Convert this into an computational graph.

$h_1 = \tanh(W_{hh} h_0 + W_{xh} x_1)$

$y_1 = W_{hy} h_1$

